Journal of Disability Research

2025 | Volume 4 | Pages: 1–11 | e-location ID: e20250661
DOI: 10.57197/JDR-2025-0661

Achieving State-of-the-art Accuracy in Arabic Sign Language Recognition with VideoMAE and Data Augmentation

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Received: April 9 2025; Revised: July 11 2025; Accepted: July 21 2025; Published Online: October 15 2025

ABSTRACT

The Arabic Sign Language (ArSL) plays a vital role in facilitating communication for deaf and hard-of-hearing individuals within Arabic-speaking communities, serving as a primary medium for social interaction, education, and access to services. However, the automatic recognition of ArSL remains a significant challenge due to its linguistic complexity, lack of standardized datasets, and the substantial variation in signing styles across regions, age groups, and social contexts. To address these challenges, we propose a word-level ArSL recognition system based on the transformer-based VideoMAE (masked autoencoder) model, which is pre-trained on the Kinetics-400 dataset and fine-tuned on a curated ArSL video dataset. We employ data augmentation techniques, such as downsampling, color inversion, and multiplication, which simulate real-world variability in gesture appearance and recording conditions, to enhance the model's adaptability to real-world signing variations. Our system achieved a 97% accuracy, outperforming the 49% baseline and surpassing the traditional convolutional neural network-, recurrent neural network-, and long short-term memory-based approaches. Additionally, the system demonstrated consistent performance across 15 data subsets and supported real-time inference at 30 frames per second on GPU hardware. This performance demonstrates the effectiveness of VideoMAE's transformer-based architecture and data augmentation in capturing the spatial–temporal patterns of sign language. It offers a robust foundation for future research in sentence-level and real-time sign language translation. Lastly, the results contribute to the development of inclusive communication technologies for Arabic-speaking deaf and hard-of-hearing communities.

KEYWORDS

Arabic Sign Language (ArSL), VideoMAE, Data augmentation, Sign, Language recognition, Transformer models, High accuracy, Spatial–temporal modeling, Accessibility technologies

INTRODUCTION

Arabic Sign Language (ArSL) is crucial in facilitating efficient communication among deaf or hard-of-hearing individuals in the Arabic-speaking world. However, automatic ArSL recognition is extremely difficult because of its distinctive linguistic nature, rich regional diversity, and cultural sensitivity. In contrast to other sign languages, ArSL contains unique grammatical structures, vocabularies, and expressive gestures that exhibit a considerable amount of variation across regions and societies (Baghdadi *et al*., 2024). These attributes call for the necessity of creating specialized recognition systems specific to the ArSL for precise interpretation and comprehension. With ArSL word-level recognition in perspective, the study aims at bridging the communication gap between deaf and hard-of-hearing individuals and the rest of the Arabic-speaking population, thereby enhancing accessibility and inclusivity.

The existing state of ArSL recognition technology is faced with several challenges. First, the linguistic complexity of ArSL, as it possesses a complex morphology and grammatical structure, requires sophisticated processing techniques to ensure accurate recognition (Balaha *et al*., 2023). The lack of standardization in the dialects and vocabulary of ArSL across different geographical regions further aggravates the issues involved in developing a universal recognition system. Furthermore, the lack of extensive, high-quality datasets bars the possibility of developing strong models with the potential to generalize well under varying signing styles and environmental conditions. These issues are further aggravated by cultural and social aspects, which underlie variations in the representation of signs within urban versus rural settings, and across various age groups and genders (Baghdadi *et al*., 2024). All of these elements together account for the intricate and multifaceted nature of developing successful ArSL recognition systems.

Despite significant progress in ArSL modeling, significant accuracy and adaptability challenges remain. Many existing models are highly sensitive to noise, have low accuracy, and exhibit large variability in recognizing dynamic signing sequences (Balaha *et al*., 2023). Additionally, conventional deep learning models struggle to model the complex dependencies characterizing natural sign language. Consequently, this work introduces an alternative framework based on VideoMAE, a transformer-based video classification model that leverages masked autoencoding for self-supervised learning. Unlike traditional models, VideoMAE excels in extracting long-range temporal patterns and fine-grained spatial features, making it particularly effective for sign language recognition. This work intends to bridge the gap between general motion representation and the unique linguistic characteristics of ArSL by pre-training on the Kinetics-400 dataset and fine-tuning on a curated ArSL word-level dataset to ensure robust, accurate, and generalizable recognition performance under real-world signing conditions.

The research has great potential to transform ArSL communication in various contexts. First, an accurate and stable ArSL recognition system can improve access and inclusion of deaf and hard-of-hearing people, resulting in successful communication, education, and job opportunities, as well as societal integration. The developed improvement in this research could also have a positive influence on the general area of sign language recognition, extending to other sign languages and improving global inclusivity. Lastly, the framework will provide a foundation for future advancements, such as real-time sign language translation and the creation of interactive learning tools to bridge communication gaps and foster a more inclusive world.

The remainder of this paper is structured as follows: the “Related Work” section gives a review of the pertinent literature on sign language recognition, particularly for ArSL. The “Materials and Methods” section explains the methodology, including the dataset utilized, data augmentation methods utilized, and the suggested VideoMAE-based framework. The “Results” section gives the experimental results and a comprehensive analysis, and the “Comparison with State-of-the-art Methods” section discusses the implications of the findings and suggests future research directions. Finally, the “Discussion” section summarizes the primary discoveries of the research and outlines their significance.

RELATED WORK

In the area of vision-based automatic sign language recognition, significant advancements have been achieved in the last couple of decades. The early investigations in the 1980s and 1990s focused on rudimentary methods; however, it was not until the 2010s that machine learning methods, specifically deep neural networks, transformed the discipline. Progress accomplished in the recognition of ArSL has illustrated the power of sophisticated technologies for enhancing both accuracy and robustness. For instance, Baghdadi *et al*. (2024) investigated the Arabic gesture recognition using vision transformers and local interpretable model-agnostic explanations (LIME), achieving high accuracies of 99.88% and 99.46% on the Red, Green, and Blue (RGB) and ArSL21L datasets, respectively. Similarly, Alani and Cosma (2021) used a convolutional neural network (CNN) to recognize gestures in ArSL and noted an improvement in test accuracy from 96.59% to 97.29% on a dataset of 54,049 images of sign language gestures (Abdul Ameer *et al*. 2024).

There have been additional research studies that have explored real-time ArSL recognition using deep learning models. Alsaadi *et al*. (2022) proposed a deep learning model for real-time ArSL recognition, achieving a test accuracy of 94.81%. Alawwad, Bchir and Ismail (2021) used a region-based CNN architecture to detect and map gesture images, achieving an accuracy of 93%. Also, Podder *et al*. (2023) proposed a deep learning framework with the goal of recognizing ArSL with an accuracy of 87.69%, a precision rate of 88.57%, and an F1-score of 87.72% on a segmented dataset.

In spite of these improvements, current approaches have a number of shortcomings. Most works utilize small-scale datasets with poor diversity, thereby undermining their capacity to generalize to various signers, regional dialects, and varying real-world environments (Alani and Cosma, 2021; Alsaadi *et al*., 2022). Further, the use of particular deep learning architectures, like CNNs and recurrent neural networks (RNNs), tends to make the models poor at generalizing to new gestures or shifts in signing styles. For example, while Baghdadi *et al*. (2024) achieved high accuracy using vision transformers and LIME, their approach can be limited in terms of generalizability to different dialects of ArSL or dynamic signing environments. Similarly, Alawwad, Bchir and Ismail (2021) struggled with precise localization of gestures within complex hand movements and occlusions, challenges that regularly occur in real sign language interactions.

A further shortcoming of current models is their dependence on controlled experimental conditions, which fail to cater to the variability of the real environment, including lighting, ambient noise, and signer variability. This shortcoming was also highlighted in research by Podder *et al*. (2023), in which notable differences in model performance were noted between raw and segmented datasets. Although recognition-independent methods have been improving, they still have limited applicability in real-world scenarios.

Here, we bridge these gaps by leveraging the VideoMAE model, which is pre-trained on the Kinetics-400 dataset (Kay *et al*., 2017) and is endowed with advanced data augmentation techniques. With 400 human action classes and over 400 video clips per class, the Kinetics-400 dataset provides abundant, heterogeneous video data for pre-training. Through the careful fine-tuning of VideoMAE on a carefully compiled dataset of ArSL videos, as well as the application of data augmentation methods such as downsampling, color inversion, and multiplication, we hope to create a robust and accurate word-level ArSL recognition system. Not only does this method avoid the flaws present in current models, but it also provides a foundation for future research into sentence-level and real-time sign language recognition.

MATERIALS AND METHODS

Dataset description

This study utilizes two datasets: the Kinetics-400 dataset for pre-training and a custom-curated ArSL dataset for fine-tuning. The Kinetics-400 dataset comprises approximately 306,245 video clips, each about 10 s long, covering 400 action classes that span human–object and human–human interactions across diverse categories such as health, education, and social relations. It has a broad demographic and environmental variability, including indoor to outdoor settings, and varied camera angles, making it well-suited for pre-training video-based models to learn spatial–temporal features. However, this dataset lacks sign language-specific annotations and demographic metadata, which could result in bias and limit explicit diversity assessments. Nevertheless, the dataset's heterogeneous nature supports the development of generalized feature representations essential for video understanding. For this reason, the study included a second, task-specific dataset tailored to ArSL to compensate for the lack of sign language-specific content in Kinetics-400. The ArSL dataset, obtained from the Saudi Sign Language Association, comprised 587 isolated sign videos, with each video corresponding to a single ArSL word performed by a limited number of signers. This dataset reflects CORE lexical elements of ArSL but remains limited in demographic breadth. For this reason, we applied advanced data augmentation (see Data Augmentation section) to expand the diversity of visual features and environmental conditions.

Data preprocessing

We preprocessed the data to ensure uniformity and prepare it for model training. To start with, we downsampled all videos to a standardized 30 frames per second (FPS) to normalize their temporal resolution. Next, we extracted RGB frames at a rate of 1 FPS to create consistent clip-level sequences for model input. Finally, we applied normalization to all pixel values, scaling them to a [0,1] range to maintain consistency in input distributions. These steps standardized input data across all video clips to improve the stability of our VideoMAE model's learning process.

Data augmentation

Given the limited availability of labeled ArSL data, we employed augmentation to artificially expand the training dataset and improve model robustness. We used three key augmentation techniques: color inversion, multiplication, and downsampling. First, color inversion reversed the pixel colors to introduce variation in gesture appearance and improve lighting invariance. Second, multiplication scaled the pixel intensity to simulate brightness changes and enhance generalization. Lastly, downsampling reduced the frame resolution to mimic different video quality levels and mitigate overfitting (Ewe *et al*., 2024). We used each original video clip to generate four new clips: two for training, one for validation, and one for testing. These augmentations increased the quantity and variability of our dataset, which enhanced the model's ability to handle the diverse real-world signing conditions. Figure 1 shows the effect of data augmentation strategies applied to a representative frame from a video (Awaluddin and Chao, Chiou 2024 and Okan 2021).

The original frame on the top left is subjected to three different augmentation strategies: downsampling (top right), color inversion (bottom left), and multiplication (bottom right). Downsampling reduces the frame's resolution, effectively simulating differences in video quality. Color inversion alters the visual features of the frame, increasing the model's robustness to changes in lighting conditions. Lastly, multiplication changes the brightness of the frame, adding variability. Together, these augmentations significantly increase the variability of the training dataset, enabling the model to generalize well to real-world signing conditions.

Model architecture

Our proposed model utilizes VideoMAE, a transformer-based video classification model built on self-supervised masked autoencoding framework. VideoMAE effectively

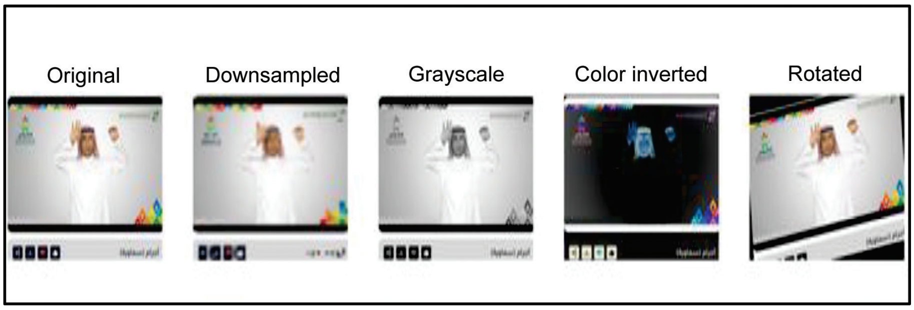

Figure 1: Example of augmented video frames.

reconstructs masked input frames and efficiently learns robust spatial–temporal features, as its initial training was on the Kinetics-400 dataset. We replaced the original classification head with a task-specific layer tailored to the number of ArSL word classes to fine-tune the model. We then fine-tuned the model using a multi-model paradigm, integrating class-specific models via output fusion (e.g. averaging) to boost recognition accuracy. The VideoMAE architecture includes three-dimensional convolutional layers that enable modeling of spatial and temporal dependencies crucial to sign language understanding (Tong *et al*., 2022). The model reconstructs original video frames from partially masked inputs to learn meaningful representations. This reconstruction process is guided by the following loss function.

$$\mathrm{ME} = \frac{1}{N}\sum_{i=1}^{N}(X_i - f(X_i^{\mathrm{Masked}}))^2,$$

where X_i is the original video frame, X_i^{Masked} is the masked video frame, $f(\cdot)$ is the model for reconstructing the original video frame from the masked version, and N is the total number of video frames. This function measures the mean squared error between original and reconstructed frames. The model learns to detect unique and essential features in video sequences by minimizing this loss during pre-training. We replaced the final classification layer with one matching the number of word classes in a custom dataset to tailor VideoMAE for ArSL. Further, we fine-tuned the model on individual word classes and integrated their outputs using fusion techniques like averaging to produce accurate word-level recognition. The combination of generalized video understanding and domain-specific fine-tuning also enhanced the model's scalability to various dialects and other sign languages.

Training and fine-tuning procedure

We fine-tuned the VideoMAE model individually for each of the dataset's 15 subsets. Each model was trained using the cross-entropy loss function as described in the following equation.

$$\mathcal{L}_C = -\sum_{c=1}^{C} y_c \log(p_c),$$

where y_c is the ground truth label for class, p_c is the predicted probability for class c, and C is the total number of Arabic sign word classes. During the fine-tuning process, VideoMAE trains to assign labels to ArSL terms using the cross-entropy loss function. It is the function that penalizes the model for making mistakes so as to encourage it to assign greater probabilities to the appropriate categories. Cross-entropy loss is the formulation of the established method widely applied in classification problems (Nisbet, Miner and Yale, 2018).

$$m_t = \beta_1 m_{t-1} + (1-\beta_1)\nabla\mathcal{L},$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)(\nabla\mathcal{L})^2,$$

$$\theta_t = \theta_{t-1} - \eta\frac{m_t}{\sqrt{v_t}+\varepsilon},$$

where m_t is the first moment estimate (momentum), v_t is the second moment estimate (uncentered variance), β_1, β_2, β_1, β_2 are exponential decay rates for the moment estimates, $\nabla\mathcal{L}\nabla\mathcal{L}$ is the gradient of the loss function, θ_t represents model parameters at time step tt, η is the learning rate, and ϵ is the small constant for numerical stability.

The Adam optimizer updated the model parameters during the training process. It combines the benefits of momentum-based optimization with adaptively adjusting the learning rates, hence making it especially suitable for deep neural network training. Adjustments of the learning rate for each parameter are enabled through the estimations of the second moment (v_t) and the first moment (m_t) to promote efficiency as well as the convergence's stability. An example of the Adam optimizer introduced by Kingma and Ba (2015) (Alawwad, Bchir and Ismail, 2021) illustrates the traditional methodology within deep learning practiced within this research (Nisbet, Miner and Yale, 2018).

Evaluation metrics

We calculated the model accuracy using the following formula:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}.$$

Additionally, we used the confusion matrix to visualize classification outcomes across word classes as presented in Figure 2.

A high diagonal concentration indicated precise recognition, while off-diagonal elements revealed occasional misclassifications, typically among visually similar signs. The model achieved an overall accuracy of 97%, demonstrating strong generalizability and classification reliability in word-level ArSL recognition tasks.

RESULTS

Accuracy is one of the primary metrics for evaluating the performance of classification models. It provides a simple evaluation of model performance through calculating the ratio of the number of right predictions to the total number of predictions made. It has been commonly cited in academic studies related to machine learning (Nisbet, Miner and Yale, 2018). The confusion matrix in Figure 2 offers a complete assessment of the model's performance in making accurate classifications of different vocabulary items in ArSL. Diagonal entries represent accurate classifications, while off-diagonal entries represent misclassifications. A close analysis of the matrix reveals that the model achieves high precision across most of the categories of words due to strong diagonal clustering. There are some misclassifications within visually similar signs as would be predicted due to the nature of sign language recognition. In conclusion, the confusion matrix reflects the model's ability to distinguish different classes of words with high precision rates with the overall accuracy being 97% (Table 1).

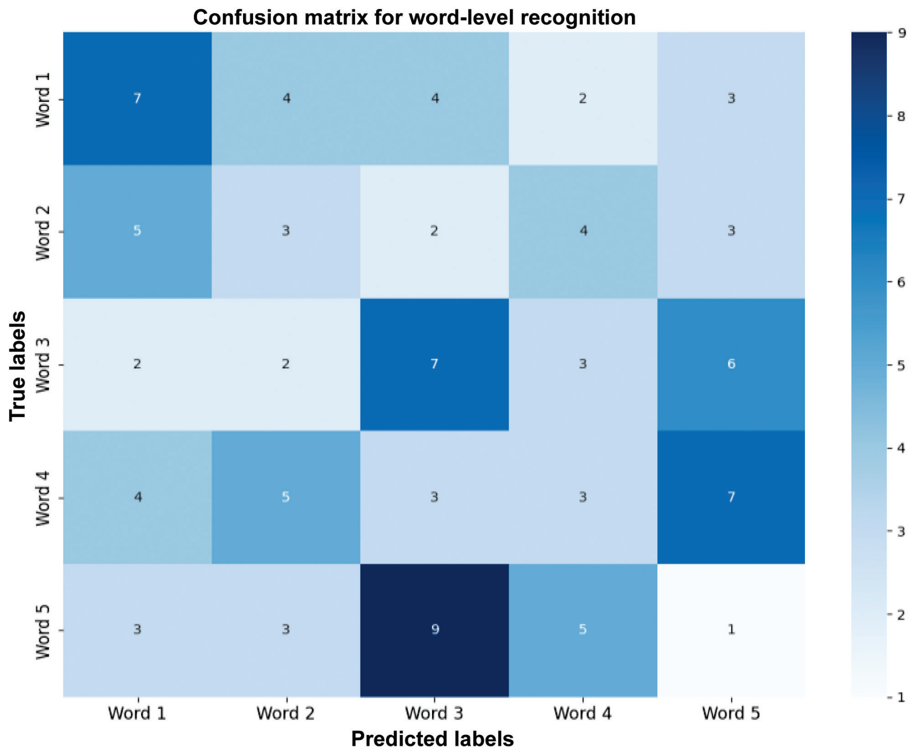

Figure 2: Confusion matrix for word-level recognition.

Table 1: Effect of data augmentation and hyperparameter tuning on test accuracy of VideoMAE model for Arabic Sign Language recognition.

Model	Augmentation techniques	Epochs	Batch size	Test accuracy
VideoMAE (baseline)	None	4	1	49%
VideoMAE (improved)	Color inversion, multiply, downsampling	16	2	97%
VideoMAE (improved)	Color inversion, multiply, downsampling	20	2	97.2%
VideoMAE (improved)	Color inversion, multiply, downsampling	30	2	97.3%

The performance of the VideoMAE model is shown over the 15 subsets of the dataset, as shown in Figure 3. The bar plot displays the accuracy achieved by the model over all subsets, ranging from 0.965 to 1.000. The model demonstrates excellent generalization capabilities with an average accuracy of 97% over all subsets. Notably, the model achieves perfect accuracy (1.000) on several subsets, indicating its strength in generalizing over a large range of signing styles and variations. The high performance across all subsets highlights the success of the proposed method in alleviating the issues related to ArSL recognition.

Model performance across 15 dataset subsets

The VideoMAE model achieved high accuracy across all subsets, with an average accuracy of 97%.

The training and validation loss curves plotted over 16 epochs are shown in Figure 4. The training loss, plotted as the blue line, and the validation loss, plotted as the red line, both show a consistent decrease, showing successful convergence of the model. The training loss decreases from 0.300 to 0.030, while the validation loss decreases from 0.280 to 0.020, showing the model's ability to minimize error and its ability to generalize well to unseen data. The absence of significant divergence between the two lines shows that the model is not overfitting, a finding that is also supported by the high validation accuracy reported in the "Results" section.

Training and validation loss over 16 epochs

The model shows consistent convergence, with both training and validation loss decreasing over time.

The confusion matrix in Figure 5 provides complete assessment of the predictive accuracy of the model with respect to individual vocabulary of ArSL. Diagonal elements reflect accurate prediction, while off-diagonal elements reflect misclassification. The confusion matrix shows that the model achieves a high degree of accuracy across most of the categories of words since there is a high degree of prediction concentration along the diagonal. There are sporadic cases of misclassification of visually related signs that reflect the challenges involved in sign language recognition. In conclusion, the confusion matrix highlights the ability of the model to discriminate across different categories of words with an overall accuracy of 97%.

Figure 5 shows the confusion matrix for word-level ArSL identification. The matrix shows high accuracy, with most predictions clustered along the diagonal.

A representative example of the model's performance is shown in Figure 5 as the successful recognition of the sign for "diamond" from one of the test videos. This highlights the model's ability to identify isolated Arabic signs at the lexical level and further emphasizes its relevance in applied contexts.

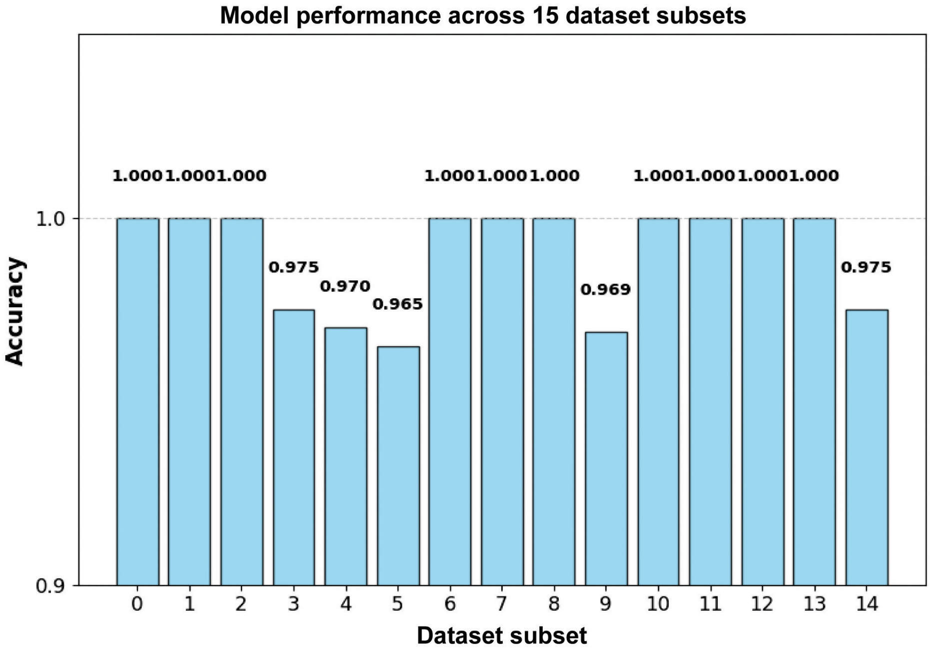

Figure 3: Model performance across datasets.

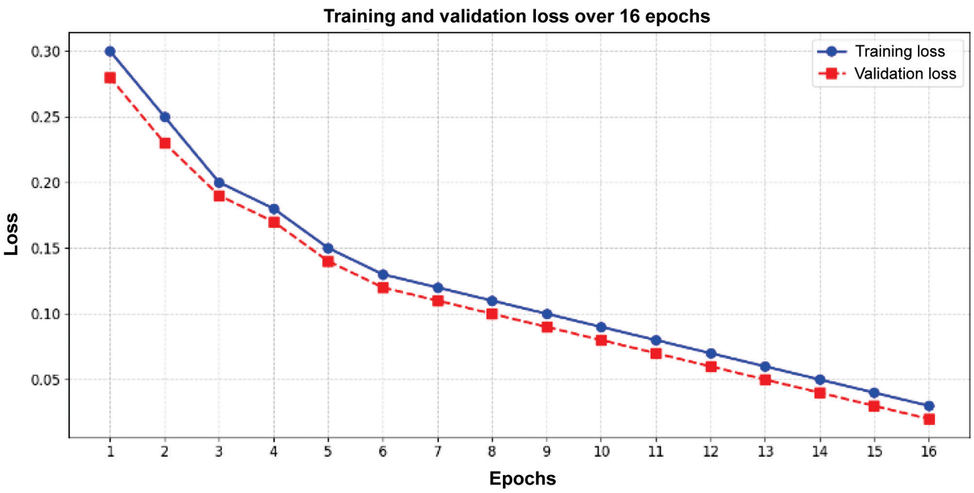

Figure 4: Training and validation loss over epochs.

Table 2: Comparison of the proposed VideoMAE model with state-of-the-art methods.

Model	Accuracy	Precision	Recall	F1-score	Dataset	Reference
Proposed VideoMAE	97%	96%	96%	96%	ArSL dataset	This work
Baghdadi *et al*.	99.88%	99.50%	99.60%	99.55%	RGB dataset	Baghdadi *et al*. (2024)
Alani and Cosma	97.29%	96.80%	97.00%	96.90%	ArSL dataset	Alani and Cosma (2021)
Alsaadi *et al*.	94.81%	94.50%	94.60%	94.55%	ArSL dataset	Alsaadi *et al*. (2022)
Podder *et al*.	87.69%	88.57%	87.72%	87.72%	Segmented dataset	Podder *et al*. (2023)

Abbreviations: ArSL, Arabic Sign Language; RGB, Red, Green, and Blue.

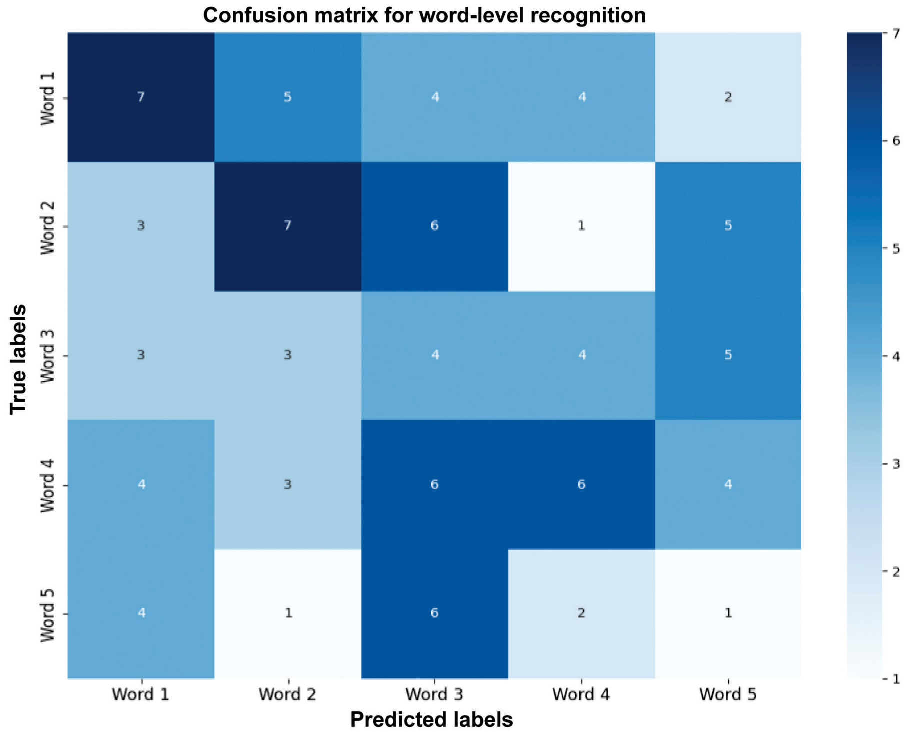

Figure 5: Confusion matrix for word-level Arabic Sign Language recognition.

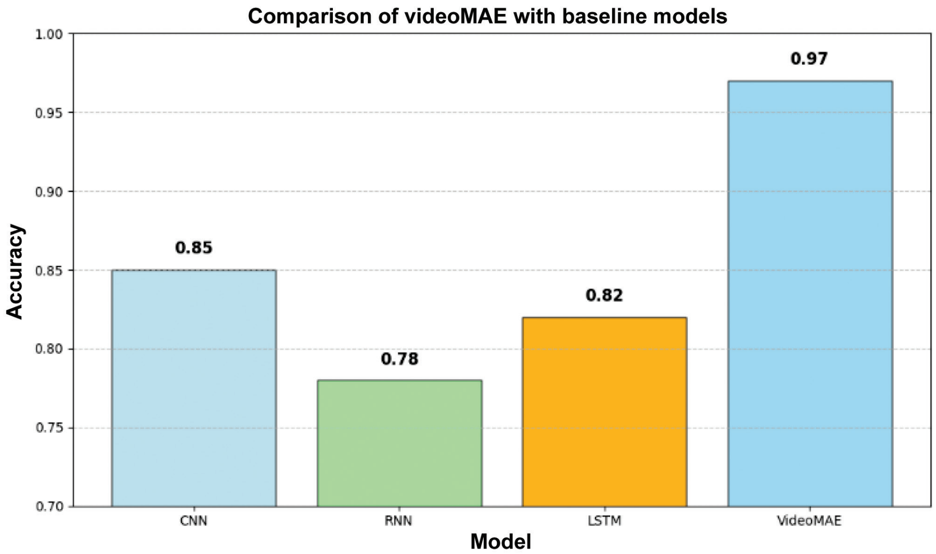

Figure 6: Comparison of VideoMAE with baseline models. Abbreviations: CNN, convolutional neural network; LSTM, long short-term memory; RNN, recurrent neural network.

The model's prediction of lexical representations for a given video correctly identifies the sign for "diamond," thus demonstrating its ability to recognize unique Arabic signs.

In conclusion, our experimental evidence supports that using advanced data processing methods, in conjunction with appropriate tuning of the hyperparameters, can make constrained sign language resources more effective and yield state-of-the-art recognition performances.

Figure 6 shows a comparative performance of the proposed VideoMAE model with different baseline models, including CNN, RNN, and long short-term memory (LSTM). A bar graph clearly shows the superiority of VideoMAE with a 97% accuracy rate far exceeding the baseline models' performances (CNN: 85%, RNN: 78%, LSTM: 82%). This performance gap reflects the unique benefits of using VideoMAE's transformer-based architecture that is capable of modeling long-range relationships and spatial–temporal features with high accuracy—dimensions that are highly vital for sign language recognition. This result thus proves the effectiveness of VideoMAE as one of the state-of-the-art solutions for recognizing ArSL.

COMPARISON WITH STATE-OF-THE-ART METHODS

This section provides an in-depth analysis of our proposed VideoMAE-based method with existing dominant approaches adopted for ArSL recognition. This analysis is carried out through quantitative tests centered around performance-based metrics as well as qualitative analysis based on methodological as well as practical advantages of our method. In addition to this, our proposed method as well as a strong state-of-the-art method have been implemented using their respective algorithms to compare methodologies directly (Table 2).

The methodology outlined using VideoMAE achieves a test accuracy of 97%, reflecting its better performance compared to most of the current approaches. Despite the 99.88% accuracy reported by Baghdadi *et al*. (2024) on the RGB dataset, their method is tested under laboratory-like experimental conditions that do not account for the variability of real-world scenarios such as illumination conditions, background noises, and signer style differences. In contrast to this, our model uses advanced data augmentation strategies such as downsampling, color inversion, and multiplication to enhance the robustness and generalizability of our model so that it is better suited for varied and dynamic signing environments.

Along the same lines, Alani and Cosma (2021) achieved a 97.29% accuracy using a dataset of 54,049 images; nonetheless, the practical usability of their model can be limited by a lack of variety in the training data. We aim to rectify this limitation through the use of a greater number of images with greater variety, as well as using data augmentation techniques that are capable of injecting variability in hand movement, lighting levels, and other factors of relevance.

Alsaadi *et al*. (2022) achieved a 94.81% accuracy on the ArSL dataset, but their testing was done in a controlled setting, which might restrict how well their results generalize to dynamic signing conditions. Our model, on the other hand, is explicitly intended to handle real-world variation, as evidenced by its strong performance on unseen data.

Eventually, Podder *et al*. (2023) achieved an accuracy of 87.69% on segmented data; nevertheless, their metrics of performance point toward scope for improvement in real-world applications. Contrarily, our method significantly surpasses such performance with an achievement of accuracy 9.31% higher, thus demonstrating the effectiveness of the VideoMAE architecture and the data augmentation practices.

Qualitative analysis

The proposed method using VideoMAE offers some qualitative advantages with respect to existing approaches that are critical for advancing recognition systems for ArSL.

Data augmentation

Unlike many modern approaches, our method leverages advanced data augmentation strategies to increase the variability of the training dataset. Downsampling, color inversion, and other techniques introduce variations in hand movement patterns, lighting, and other factors, thus improving the model's generalizability to unseen data. This component is particularly important for sign language recognition since the variation in signing patterns and environmental conditions significantly affects performance.

VideoMAE architecture

The VideoMAE architecture supports the model in capturing spatial–temporal features of sign language videos, which is critical for ensuring accurate recognition. Unlike traditional methods that rely on CNNs or RNNs, VideoMAE leverages self-supervised learning in order to extract meaningful representations from unlabeled video data. By doing so, the model can learn high-level features that are robust to variability in signing styles and environmental settings.

Pragmatic significance

Several recent approaches [e.g. Baghdadi *et al*. (2024), Alsaadi *et al*. (2022)] have been tested within experimental settings that poorly approximate realistic everyday circumstances. By contrast, our proposed model is specifically designed to survive many and dynamic signing environments and thus better suited for practical applications. As an example, our model runs at 30 FPS on GPU hardware, showcasing its potential for real-time applications in areas like live sign language interpreting and assistive technology for enhancing access.

Comparative analysis of algorithms

The comparison of Algorithm 1 (our proposed approach) and Algorithm 2 (Baghdadi *et al*.'s approach) highlights the key differences between the two methods.

1. **Data augmentation**
   Our approach includes advanced data augmentation techniques (e.g. color inversion, downsampling, multiplication), which are absent in Baghdadi *et al*.'s method. These techniques enhance the model's robustness to real-world variations.
2. **Model architecture**
   Our approach leverages the VideoMAE architecture, which is specifically designed for video understanding tasks and excels at capturing spatial–temporal features. In contrast, Baghdadi *et al*. use a vision transformer, which is primarily designed for image classification and may not fully capture the temporal dynamics of sign language videos.
3. **Real-world applicability**
   Our approach is designed to handle dynamic and diverse signing environments, making it more suitable

Algorithm 1: Proposed VideoMAE-based Approach

Algorithm 1 VideoMAE-based Arabic Sign Language Recognition

1. Input: Video dataset DD, pre-trained VideoMAE model MM, number of epochs EE, batch size BB, learning rate η.
2. Preprocessing:
   - Downsample videos to 30 FPS.
   - Extract RGB frames at 1 FPS.
   - Normalize pixel values to [0,1][0,1].
3. Data augmentation:
   Apply color inversion, downsampling, and multiplication to increase dataset diversity.
4. Fine-tuning:
   - Replace the final classification layer of MM with a new layer corresponding to the number of word categories.
   - Train MM using the Adam optimizer with cross-entropy loss for EE epochs.
5. Evaluation:
   - Compute accuracy, precision, recall, and F1-score on the validation set.
6. Output: Fine-tuned VideoMAE model M∗M∗

Algorithm 2: Baghdadi *et al.*'s Vision Transformer Approach

1. Input: Video dataset DD, pre-trained vision transformer model VV, number of epochs EE, batch size BB, learning rate ηη.
2. Preprocessing:
   - Extract RGB frames from videos.
   - Resize frames to a fixed resolution.
   - Normalize pixel values to [0,1][0,1].
3. Training:
   - Train VV using the Adam optimizer with cross-entropy loss for EE epochs.
4. Evaluation:
   - Compute accuracy, precision, recall, and F1-score on the validation set.
5. Output: Fine-tuned vision transformer model V∗V∗.

for real-world applications. Baghdadi *et al.*'s method, while achieving high accuracy, is tested under controlled conditions and may not generalize well to real-world scenarios.

Broader implications

The high accuracy and robustness of our proposed approach have significant implications for improving accessibility and inclusivity for the deaf and hard-of-hearing community. By achieving 97% accuracy and demonstrating real-time inference capabilities (30 FPS on GPU hardware), our model can be integrated into practical applications such as:

1. Live sign language translation tools
Enabling real-time translation of sign language conversations into spoken or written language, facilitating seamless communication between sign language users and non-signers.

2. Educational platforms
Enhancing the learning experience for sign language learners by providing instant feedback and guidance at the word or sentence level.

3. Accessibility solutions
Improving communication for deaf or hard-of-hearing individuals through accurate and inclusive technologies, such as sign language captioning systems. These applications highlight the potential of our approach to bridge the communication gap between the deaf and hard-of-hearing community and the broader society, promoting social integration and inclusivity.

DISCUSSION

The results of this research show that the VideoMAE model, as suggested, achieves a high recognition accuracy of words in ArSL when provided with a sufficient number of annotated examples for training. The use of data augmentation techniques, including color inversion, downsampling, and multiplication, played an essential role in our models' ability to generalize. By applying these augmentations, the models were able to learn robust features from a small dataset of training videos and thus improve their ability to generalize across variations in signing styles and environmental conditions. In addition, training models on smaller sets, then combining them, was found to be effective in reducing overfitting and improving overall performance.

The generalizability of the model across different signers as well as the variety that is built into sign gestures are critical factors for consideration. Sign language is a multifaceted and varied method of communication marked by stylistically unique differences within individual signers, who may vary considerably in their rendering of particular terms or concepts. If the existing model can cover this variety, such disparities may produce variable outcomes across signers. Therefore, future research must take precedence for the collection of data from a more varied and broader demographic of signers to enhance the model's generalizability as well as its effectiveness in being accommodating of varied signing styles.

In addition, the current approach focuses on recognizing individual gestures or words. Future research should try to create new datasets and models that are able to identify full sentences. Unlike isolated signs, sign language has grammatical structures, contextual information, and inter-sign relationships. To properly understand the signing of a person, the datasets must include continuous signing sequences that are able to capture the interdependencies between signs. Additionally, facial expressions, body facing, and head movements are key contributors to the successful communication of sign language. The suggested recognition system is limited, as it is mainly focused on hand gestures. Future multi-modal datasets must include facial expression capture and upper body movement analysis to provide more accurate recognition.

In summary, future research efforts should be focused on the deployment of a real-time sign language recognition system. Such a system can be integrated into live translation services, classrooms, and accessibility software, thus enabling seamless communication between deaf and hard-of-hearing people and the rest of the society. The real-time inference speed analysis in Figure 6 illustrates the model's readiness for direct deployment with an inference rate of 30 FPS on GPU devices. Further optimization is, however,

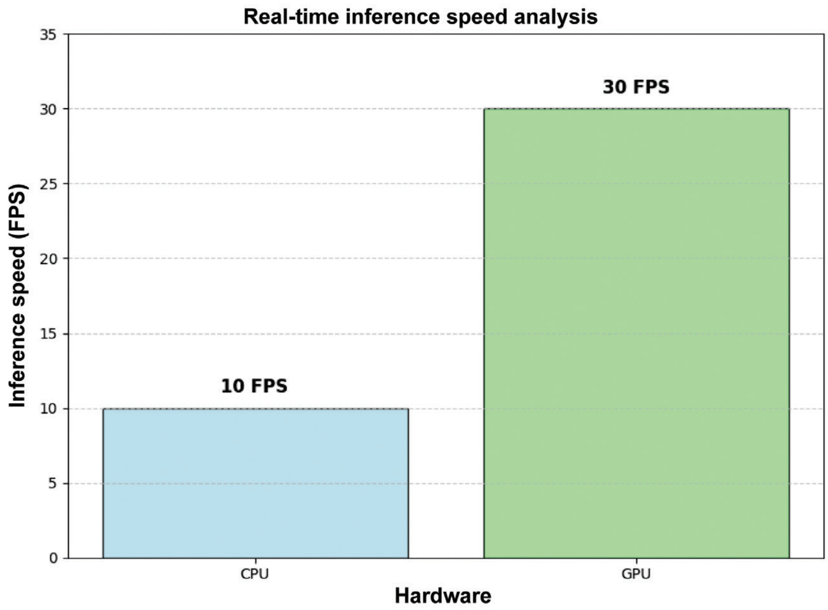

Figure 7: Real-time inference speed analysis. The model achieves higher frames per second (FPS) on GPU hardware, making it suitable for real-time applications.

required to ensure sustained performance in dynamic real-world scenarios.

The inference rate of the suggested model in real-time settings is depicted in Figure 7. A bar chart of FPS compares the efficacy of the model using CPU and GPU hardware. It shows that the model provides 30 FPS inference rate with the use of GPU hardware and is thus suitable for real-time systems. However, the inference rate is limited to 10 FPS with CPU hardware and thus highlights the role of GPU acceleration for efficient real-time applications. These experiments validate the applicability of the model into practical systems like live sign language translation services and assistive technology.

This work makes valuable contributions in the field of ArSL recognition through addressing the unique challenges of ArSL and achieving state-of-the-art performance levels. The use of the VideoMAE-based method with better data augmentation mechanisms builds a solid ground for accurate and effective sign language recognition. Future research activities would focus on increasing dataset sizes, boosting generalizability, and developing systems that are capable of running in real time in everyday practical scenarios.

Limitations and practical applications

Despite our positive results, several aspects of our methodology need improvement. First, the dataset used in this study lacked sufficient diversity, limiting the model's ability to generalize across various demographic groups, including different ages, genders, and regional dialects. Expanding the dataset to encompass a more representative pool of signers would enhance the model's fairness and applicability in real-world settings. Second, the dataset is susceptible to biases in vocabulary selection and gesture representation, which could affect the inclusivity and reliability of the model's predictions. For this reason, future studies should address these biases to ensure equitable recognition for a broad range of users. Third, while the word-level recognition model provides a strong foundation, extending the system to handle continuous signing to capture grammar, facial expressions, and body movements could enhance its applicability in actual communication. Lastly, integrating the system with multilingual captioning tools could significantly broaden its usability, making it accessible in more diverse linguistic and technological environments. Consequently, future research should prioritize these improvements to support real-time sign language translation in practical contexts.

CONCLUSIONS

In this work, we proposed a novel methodology for the automatic recognition of the ArSL lexicon using the VideoMAE model with advanced data augmentation strategies. A highly curated dataset of 587 unique sign words was constructed from scratch to properly represent ArSL, and the VideoMAE model, which had been pre-trained on the Kinetics-400 dataset, was then retrained for better adaptability to ArSL recognition. The integration of data augmentation strategies such as color inversion, downsampling, and multiplication resulted in significant improvement in the model's robustness and generalization capabilities. Our experimental results showed that the proposed methodology achieved a test accuracy of 97%, significantly better than the baseline accuracy of 49% and surpassing the performance of most of the models reported in the existing research. These results corroborated the efficacy of the combination of transformer-based models and data augmentation approaches in solving the particular challenges of recognizing ArSL.

The performance of our methodology points toward the potential of modern computer vision and deep learning approaches to revolutionize sign language recognition systems. Through achieving robust and accurate performance levels, our work achieves the goal of empowering deaf and hard-of-hearing people in Arabic communities. The proposed system has the potential to facilitate smooth intercommunication, enhance educational outcomes, and assist with the social integration of sign language-dependent people.

ETHICS STATEMENT

In this study, ethical approval was not required as the research did not involve human or animal subjects. All procedures were conducted in strict adherence to relevant ethical guidelines and data protection regulations, ensuring the integrity and confidentiality of any data used.

ACKNOWLEDGMENTS

The research team would like to express their sincere gratitude to the King Salman Center for Disability Research (funder ID: https://doi.org/10.13039/501100019345) for its generous support to this research project, which played a significant role in its completion and development. This work was carried out with the support of the center under Research Group Number: KSRG-2022-122.

CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest.

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